

Q-1

```
saksham@SAKSHAMS-PC:~/ns3/ns-allinone-3.41/ns-3.41$ ./ns3 run "scratch/routing"
Consolidate compiler generated dependencies of target scratch_routing
--- Starting Simulation with RIP (Distance Vector) ---
=====
Results for RIP
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.446 ms
Total Packets Received: 177278
=====
--- Starting Simulation with Global Routing (Link State) ---
=====
Results for Global Routing
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 208.146 ms
Total Packets Received: 177279
=====
saksham@SAKSHAMS-PC:~/ns3/ns-allinone-3.41/ns-3.41$ |
```

Q-2

```
Consolidate compiler generated dependencies of target scratch_routing
--- Starting Simulation with RIP (Distance Vector) ---
--- LINK FAILURE EVENT at 20s ---
=====
Results for RIP
Packet Delivery Ratio: 41.7496%
Average End-to-End Delay: 211.787 ms
Total Packets Received: 75252
=====
--- Starting Simulation with Global Routing (Link State) ---
--- LINK FAILURE EVENT at 20s ---
=====
Results for Global Routing
Packet Delivery Ratio: 6.16565%
Average End-to-End Delay: 181.66 ms
Total Packets Received: 11252
=====
```

Q-3

Time Instant	Gateway (Global Routing Protocol)	Gateway (RIP)
10 sec	10.0.2.2	10.0.2.2
60 sec	10.0.2.2	10.0.2.2
120 sec	10.0.2.2	10.0.2.2
180 sec	10.0.2.2	10.0.2.2
240 sec	10.0.2.2	10.0.7.2
300 sec	10.0.2.2	10.0.7.2

Q-3: Why RIP is a dynamic protocol ?

Ans-3 From 10s → 180s, the gateway is 10.0.2.2

After the link failure propagates (around 240s), RIP automatically switches the route to 10.0.7.2

This change shows that RIP continuously exchanges distance-vector updates and recomputes routes when topology changes.

When the N2–N3 link fails, RIP receives updated distances from neighbors, detects that the path via 10.0.2.2 is no longer valid, and selects an alternate path using the N2 → N7 → N6 → N3 chain.

Therefore, RIP is dynamic because it automatically adapts to link failures and updates its routing table without restarting the simulation.

Q-3 Explain the difference in packet delivery ratio(PDR) reported in question 2 based on the routing table.?

Ans-3 RIP maintains a higher PDR because it dynamically recomputes routes after the link failure, whereas Global Routing suffers a low PDR because it does not recompute routes and continues forwarding packets along a failed path.

RIP updates its routing table and reroutes traffic through gateway **10.0.7.2** once the original path via 10.0.2.2 is broken, packets continue flowing, and the PDR remains high.

Global Routing computes all paths only once at the beginning of the simulation.

When the N2–N3 link fails, the next-hop for 10.0.3.2 incorrectly remains 10.0.2.2 at all timestamps ;10s → 300s: still 10.0.2.2

Q-4

```
saksham@SAKSHAMS-PC:~/ns3/ns-allinone-3.41/ns-3.41$ cmake --build build -j$(nproc)
./ns3 run "scratch/routing"
[ 0%] Building CXX object scratch/CMakeFiles/scratch_routing.dir/routing.cc.o
[ 0%] Linking CXX executable ns3.41-routing-default
Consolidate compiler generated dependencies of target scratch_routing
--- Starting Simulation with RIP (Distance Vector) ---
--- LINK FAILURE EVENT at 20s ---
=====
Results for RIP
Packet Delivery Ratio: 97.2096%
Average End-to-End Delay: 220.347 ms
Total Packets Received: 176736
=====
--- Starting Simulation with Global Routing (Link State) ---
--- LINK FAILURE EVENT at 20s ---
=====
Results for Global Routing
Packet Delivery Ratio: 97.2103%
Average End-to-End Delay: 219.832 ms
Total Packets Received: 177273
=====
saksham@SAKSHAMS-PC:~/ns3/ns-allinone-3.41/ns-3.41$
```

Q-4 Explanation

In Q4, the routing tables contain additional Destination entries (new networks 10.0.8.0, 10.0.9.0, 10.0.10.0). This shows that both routing protocols discovered the redundant path we introduced. Although the Gateway field for the main route does not change significantly, the presence of these new destination networks is evidence that the alternative path is part of the routing database. As a result, when the N2–N3 link fails at 20 seconds, both RIP and Global Routing retain connectivity using the 3–7–8–6 path, leading to identical high packet delivery ratios.(~97.2%)